

Operational systems and data custody

A map of operational registers, their read-only reporting copies, and the handoffs that must be understood before interpreting extracts. This draft is prepared for data and operations leads as part of Mesa Lantern Energy's initial information handoff. It describes the modeled operation at the stated date and should be read with the period and coverage of its supporting records.

Purpose and immediate use

Each system has a business role. Field Log captures readings, Meter Historian contributes meter observations, Production Ledger holds reviewed volumes, and the maintenance and inspection registers preserve their own decisions. Reporting Mart receives exports and is not an editing surface. A diagram should show direction of transfer and where a person reviews an exception.

System roles and handoffs

Field Log and Meter Historian feed the reviewed Production Ledger. Inspection Register and Work Register exchange references when a finding leads to maintenance. Vendor Ledger carries contractor transactions. Reporting Mart receives read-only exports for monthly summaries; it does not write changes back to operational systems. Source entries and timestamped revisions remain in those systems, which is why a report should name its extraction date.

Daily field-to-ledger path

Field operators enter well readings in Field Log by the next morning. Production Accounting compares those entries with Meter Historian readings, records exceptions, and posts reviewed values to Production Ledger. A correction preserves the first entry and a reason for the change. The review path matters because a later extract should be traceable to the reading and decision that produced it.

Monthly close boundary

Production Accounting freezes the monthly reporting snapshot on the tenth business day after month end. Changes after that point use a dated adjustment record rather than overwriting the frozen view. A reader comparing two months must therefore distinguish the original close from a later restatement view. The reporting period and extraction date are both necessary to interpret a number.

Record linkage

Synthetic well, asset, location, vendor, work-order, and project keys connect the proposed datasets. Business users should see natural asset and project names where possible, while the keys preserve joins across extracts. A linked inspection and work order should refer to the same asset instance and location. A record without a valid link should be flagged for review rather than attached to the nearest plausible item.

Coverage and missing values

The proposed extracts end on August 31, 2026. Production rows represent selected well-days, about 32 of 118 modeled wells per day on average, so their sums are not companywide production totals. Some meter or downtime values may be null, with a reason where available. A missing value is excluded from a numeric calculation and must never be silently treated as zero.

What the client extract can show

The proposed files are selective extracts from modeled operational registers through August 31, 2026. Reporting Mart can bring those copies together without changing a source system. The package can explain definitions, ownership, and recorded activity within that window. It cannot establish complete history, unrecorded field conditions, or companywide production from a selected well-day sample.

Record question for review

Trace a corrected production value across Field Log, the meter comparison, Production Ledger, and the reporting copy. Write down the extraction time for each view. A discrepancy may be a timing difference, an unresolved review, or a broken transfer; the diagram alone cannot decide which.

Review action and handoff

Request a sample source-to-report trace for a corrected reading and a linked inspection work order. If the records disagree, keep both versions visible until their owners explain timing. The reporting copy may lag a source revision, so an extraction timestamp belongs beside any comparison.